Nhat Minh Doan (Đoàn Nhật Minh)

Affiliation

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Research area

Hyperbolic geometry and low-dimensional topology. My current research concerns the multiplicity of simple closed geodesics on hyperbolic surfaces.

Education

PhD in Mathematics, University of Luxembourg, Luxembourg — 2018 - 2022
Advisor: Prof. Hugo Parlier
Thesis: Quantifying some properties of curves and arcs on hyperbolic surfaces

MSc in Mathematics, University of Nice Sophia Antipolis, France — 2016 - 2017
Advisor: Prof. François Labourie
Thesis: Arithmetic and ergodic theory of hyperbolic surfaces

BA in Mathematics, University of Education, Ho Chi Minh City, Vietnam — 2013 - 2015

Academic positions

[3] **Researcher**, Institute of Mathematics, VAST, Vietnam — 2022 - present

[2] **SASEA Research Fellow**, National University of Singapore, Singapore — 2024 - 2026
Mentor: Prof. Ser Peow Tan

[1] **Doctoral Researcher**, University of Luxembourg, Luxembourg — 2018 - 2022

Papers

[11] Density of length ratios on Teichmüller space — 2026
(with Hanh Vo)
arXiv:2610.24700.

[10] Stable-norm boundary geometry on once-punctured tori — 2026
(with Xiaobin Li and Van Nguyen)
arXiv:2605.14574. Submitted.

[9] Graded Bridgeman dilogarithm identities on hyperbolic surfaces — 2026
(with Ara Basmajian, Hugo Parlier and Ser Peow Tan)
arXiv:2601.03876. Submitted.

[8] Optimal Farey sequence for the congruence subgroup $\Gamma_0(2^n)$ — 2026
(with Sang-hyun Kim, Mong Lung Lang and Ser Peow Tan)
arXiv:2601.01324. Submitted.

[7] Some arithmetic aspects of ortho-integral surfaces — 2026
(with Khanh Le)
Trans. AMS, 379 (2026), 6365 - 6392. arXiv:2504.09403

[6] Some remarks on subgroups of free-by-cyclic groups — 2026
(with Hoang Thanh Nguyen)
Vietnam J. Math., 54, 715 - 728 (2026).

[5] Self-intersection of arcs on a pair of pants — 2025
(with Hanh Vo)
Bull. LMS, 57 (2025), 2190 - 2209. arXiv:2407.17750

[4] Optimal special polygons for the congruence subgroups $\Gamma_0(p)$ and $\Gamma_0(pq)$ — 2025
(with Sang-hyun Kim, Mong Lung Lang and Ser Peow Tan)
J. Geom. Anal., Volume 35, 216 (2025). arXiv:2209.13937

[3] Ortho-integral surfaces — 2025
Adv. Math., Volume 466, April 2025, 110162. arXiv:2112.10694

[2] Geometric filling curves on punctured surfaces — 2023
Glasgow Math. J., Volume 65, Issue 2, May 2023, pp. 383 - 400. arXiv:2012.02461

[1] Measuring pants — 2023
(with Hugo Parlier and Ser Peow Tan)
Trans. AMS, 376 (2023), 5281 - 5306. arXiv:2002.02738

Other writings

[1] Almost all simple lengths are unique up to isometry — 2026
hexagon:2610.00298v1.

[2] The torus with asymptotically the fewest simple closed geodesics — 2026
(with Phuc Thinh Dang and Trong Toan Dao)
hexagon:2610.00159v1.

Referee work

International Mathematics Research Notices; Proceedings of the American Mathematical Society; Mathematische Annalen; Transactions of the American Mathematical Society; Conformal Geometry and Dynamics.

Talks

[28] The McShane-Rivin norm ball and some applications — Aug-Sep 2026
Remarkable Developments on Hyperbolic Geometry, Hebei Normal University, Shijiazhuang, China

[27] Seminar, TIMAS, Thang Long University, Vietnam — June 2026

[26] Topology and Geometry Seminar, NUS, Singapore — May 2026

[25] Topology and Geometry Seminar, Southwest Jiaotong University, China — March 2026

[24] Workshop on Geometry and related topics, Phu Tho, Vietnam — Jul 2025

[23] Grad Student Seminar, NUS, Singapore — Apr 2025

[22] Teichmüller Theory and Flat Structures, Matrix, Creswick, Australia — Mar 2025

[21] Workshop on Dynamics and Related Topics, IMS, NUS, Singapore — Mar 2025

[20] Workshop on Geometry and Dynamics in Low Dimensions, IMVAST, Hanoi — Jan 2025

[19] Scientific workshop for young researchers, Vinh Phuc, Vietnam — Sep 2024

[18] IMVAST-OCAMI Joint Conference, Osaka Metropolitan University, Japan — May 2024

[17] Geometry and Topology Seminar, KIAS, Korea — Mar 2024

[16] Workshop on Hyperbolic Geometry of Numbers, HCMC, KIAS, Korea — Dec 2023

[15] JSPS-VAST Workshop, FPT University Quy Nhon, Vietnam — Sep 2023

[14] Mathematics workshop and related fields, Ha Tinh University, Vietnam — Apr 2023

[13] JSPS-VAST Bilateral Joint Research Project Workshop, IMVAST — Dec 2022

[12] Algebra and Number Theory Seminar, IMVAST — Nov 2022

[11] Scientific conference on Geometry, Analysis and Applications, Phu Tho, Vietnam — Oct 2022

[10] Topology and Geometry Seminar, NUS, Singapore — Sep 2022

[9] Scientific workshop for young researchers, Vinh Phuc, Vietnam — Aug 2022

[8] Japan-Vietnam Singularity Seminar (Online), IMVAST — Apr 2022

[7] Mostly Teichmüller Spaces Seminar (Online), Queens College, USA — Mar 2022

[6] Ph.D. Defense, Geometry Winter Workshop, Luxembourg — Jan 2022

[5] Topology and Geometry Seminar (Online), IMVAST — Nov 2021

[4] Topology and Geometry Seminar (Online), VNU, Ho Chi Minh City, Vietnam — Oct 2021

	[3] PhD Away Days (Online), University of Luxembourg, Luxembourg	Oct 2020
	[2] PhD Seminar (Online), University of Luxembourg, Luxembourg	May 2020
	[1] School on Geometry and Dynamics on Hyperbolic Surfaces, Switzerland	Aug 2018
Services	[3] Co-organizing Geometry and Dynamics in Low Dimensions Workshop, Hanoi With S. Kim, T. Koberda, C. Leininger, S. P. Tan, T. T. Nguyen, T. D. Do, and V. T. Tran	Jan 2025
	[2] Co-organizing Geometry, Topology & Dynamics Seminar, National University of Singapore With Tengren Zhang and Daren Wei	2024 - 2026
	[1] Co-writing a short comic story: Maths, Magic and Maryam	2020
Advising	[3] Undergraduate student: Phuc Thinh Dang	2026 - present
	[2] PhD student: Van Nguyen	2025 - present
	[1] Vietnamese REU students: Trong Toan Dao and Anh Tuan Nguyen Co-supervised with Ser Peow Tan	2023 - 2024
Teaching	[5] Modern Geometry, Master program, Institute of Mathematics, VAST	2023
	[4] Probability, Applied Mathematics and Data Science students, Vietnam France University	2022
	[3] Linear Algebra (teaching assistant), University of Engineering and Technology	2022
	[2] Analysis for Applications (teaching assistant), University of Luxembourg	2018 - 2020
	[1] Elementary Mathematics, children aged 5-10, Mathnasium Vietnam	2012 - 2013
Awards	[1] Excellent Doctoral Thesis Award, University of Luxembourg	2022
Grants	[5] Singapore Academies South-East Asia Fellowship, hosted at the National University of Singapore	2024 - 2026
	[4] NAFOSTED, Vietnam (2 years)	2024 - 2026
	[3] Research Grant, Institute of Mathematics, VAST, Vietnam (2 years)	2022 - 2024
	[2] PRIDE Grant, Fonds National de la Recherche, Luxembourg (4 years)	2018 - 2022
	[1] Master 2 Excellence Scholarship (10 months) J. A. Dieudonné Mathematics Laboratory, Nice, France	2016 - 2017

References

Prof. Ara Basmajian
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Prof. François Labourie
EPFL, Institute of Mathematics
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Prof. Hugo Parlier
University of Fribourg
hugo.parlier@unifr.ch

Prof. Ser Peow Tan
National University of Singapore
mattansp@nus.edu.sg

Hobbies Table tennis, badminton, hiking